CII National Kaizen Circle Compétition 2019

A Kaizen Presentation By

EVEREST INDUSTRIES LIMITED

(Podanur Works, Coimbatore)

On: 11th & 12th June 2019

Venue : India Habitat Centre , New Delhi

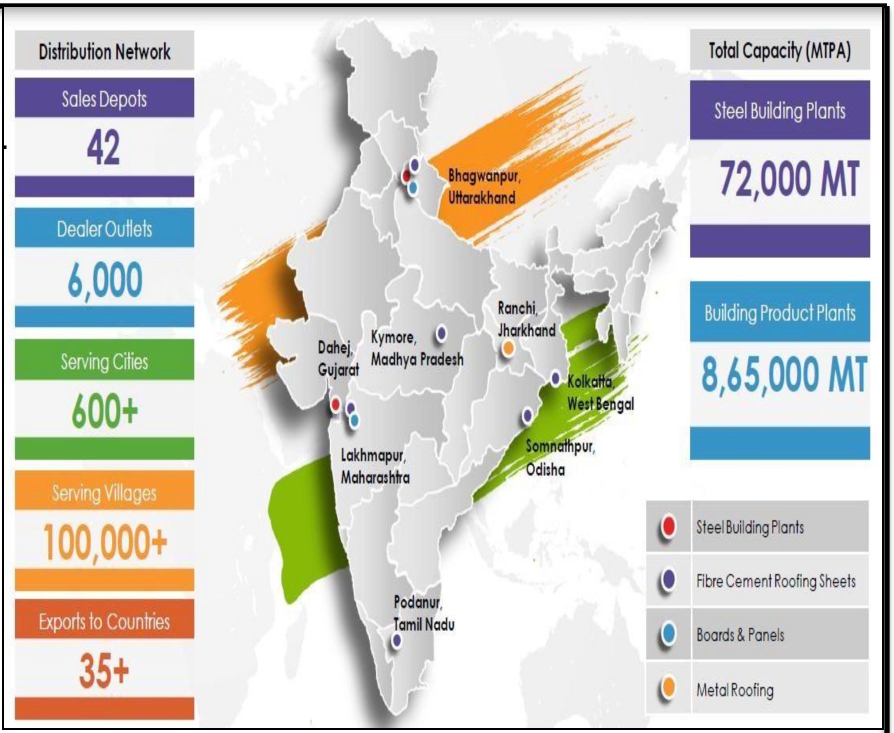
Company Profile

Company Overview:

Everest Industries Limited, incorporated in 1934, has a rich history in manufacturing of Building products and Steel products. Everest offers a complete range of roofing ceiling, wall, flooring and cladding products distributed through a large network, and also pre-engineered steel buildings for industrial, commercial and residential applications. It is one of the leading building solution providers in India, providing detailed technical assistance in the form of design, drawings and implementation for every project.

Our Business:

- **Building Products (63%) –** include fiber cement roofing sheets, fiber cement boards, solid wall panels.
- **Steel buildings (37%) –** offers customized building solutions like Pre-Engineered Steel buildings, metal roofing sheets and cladding.




Fibre Cement Roofing Sheets


Hi-Tech Roofing Sheets


Super Sheets


Cement Planks


Fibre Cement Boards


Rapicon Walls


Smart Steel Buildings

Kaizen Theme

Kaizen Theme: To reduce Power Consumption of Main Mixer Pump Motor.

Implemented Area: Sheeting Machine-1 Shop Floor, Podanur Works, Coimbatore

Implementation Date: 10th December 2018

Cross Functional Team:

Mr. M. Vijay Kumar
(Sr. Engineer Electrical Maintenance)

Team Leader
Mr. K. Karthikeyan
(Sr. Engineer Maintenance)

Mr. K. Senthil Kumar
(Electrical Supervisor)

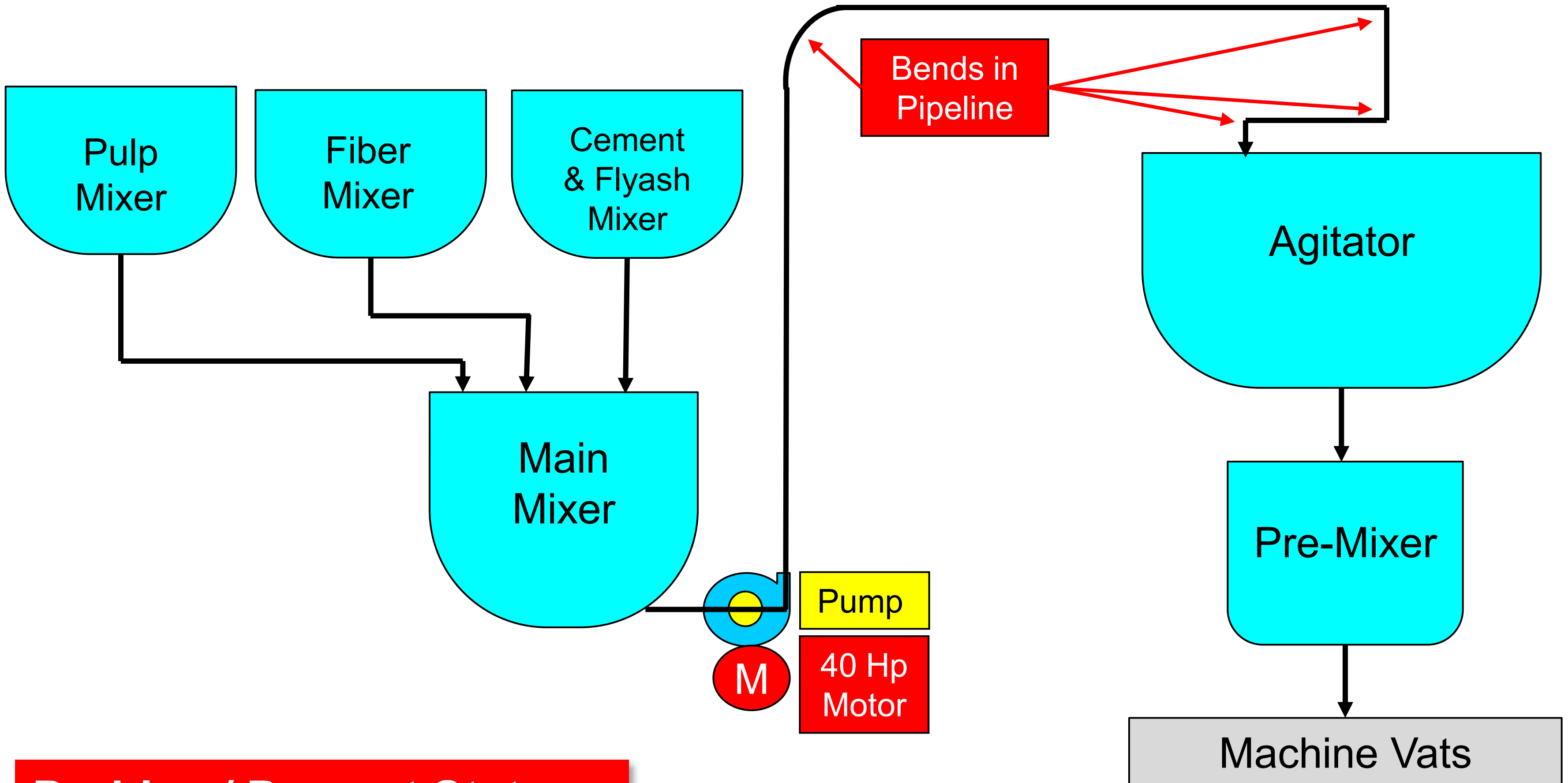
Mr. M. Vinoth Kumar
(Engineer - Production)

Mr. N. Anil Kumar
(Engineer - QS & BE)

Mr. Sasidharan
(Fitter)

Problem / Present Status

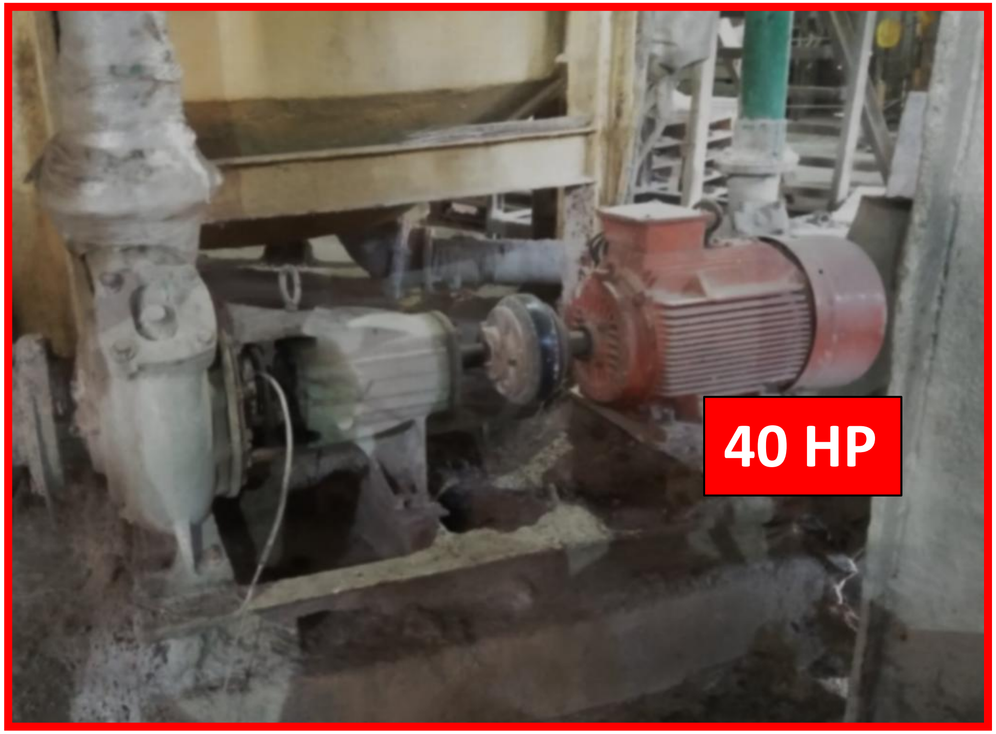
Process Flow:



Problem/ Present Status :

To reduce Power consumption of Main Mixture Pump Motor.

Problem / Present Status



Problem/ Present Status :

To reduce Power consumption of Main Mixture Pump Motor.

Root Cause Analysis & Idea Generation

Problem: More Power consumption by Main Mixer Pump motor.

Why 1 : We are using High capacity Pump Motor of 40 Hp.

Why 2 : Discharge pipe line routing has 4 Bends in b/w & length of discharge line is more.

Why 3 : Design Constraint

Root Cause: Design constraint of the Pipe line system & High Capacity Discharge Pump Motor(40 HP).

Idea Generation: To modify pipeline routing arrangement for easy flow of material in pipeline & to replace 40Hp.

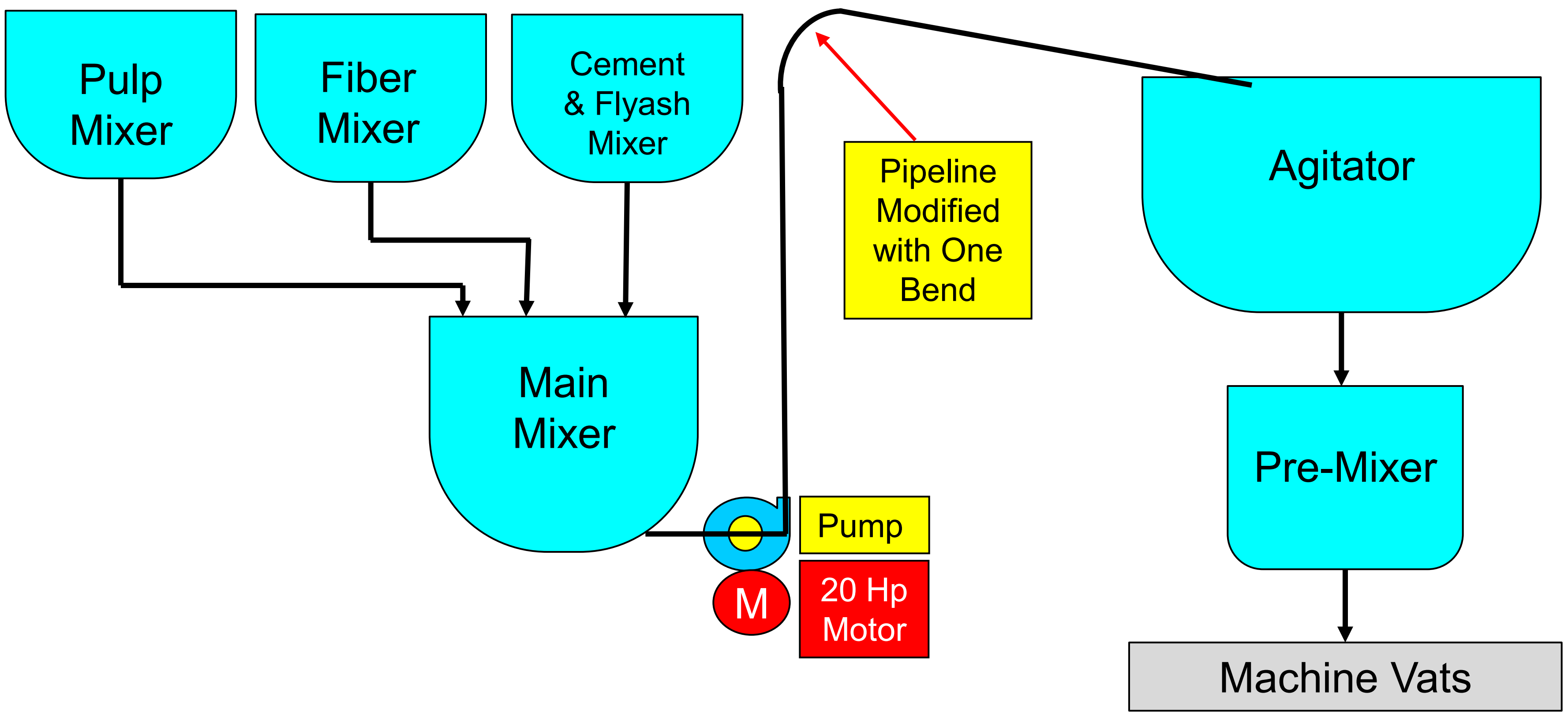
Action: Modified Pipeline routing.

Countermeasure

SN	Counter Measure Activities	Target Date	Responsibility	Status
01	Distance of the Discharge pipeline is reduced by providing direct Pipeline connection with One bend in the pipeline.	08/12/2018	Mr. K. Karthikeyan & Mr. Sasidharan	Completed
02	Replaced 40 Hp motor with 20 Hp motor.	10/12/2018	Mr. K. Vijay Kumar & K. Senthil Kumar	Completed

EVEREST

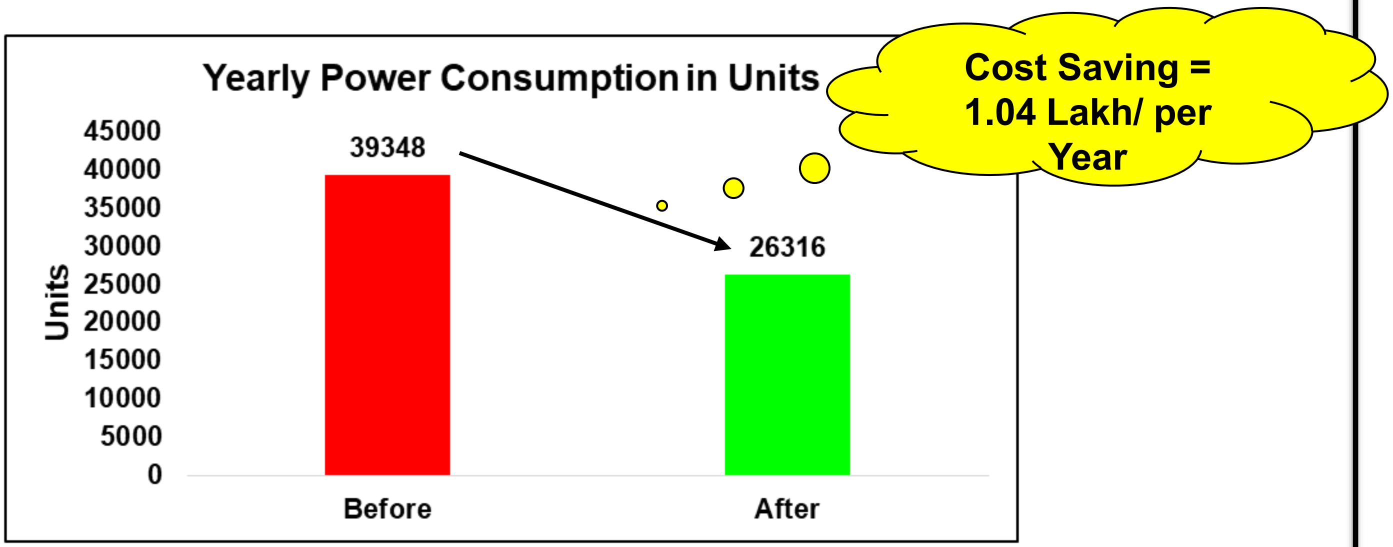
After Situation



- Distance of the Discharge pipeline is reduced by providing Direct Pipeline connection with One bend in the pipeline.
- And replaced 40 Hp Motor with 20 Hp Motor.

Validation of Data: Calculation

Before (40 Hp Motor)		After (20 Hp Motor)	
Amperes	62 A	Amperes	22 A
Average Batches in a Day	150 Batches	Average Batches in a Day	150 Batches
Power Consumption formulae	$\sqrt{3} \times V \times I \times \cos\phi$	Power Consumption formulae	$\sqrt{3} \times V \times I \times \cos\phi$
Power Consumption	40.11 KW	Power Consumption	13.44 KW
Monthly Power Consumption	3279 Units	Monthly Power Consumption	2193 Units
Yearly Power Consumption	39348 Units	Yearly Power Consumption	26316 Units
Monthly Cost Consumption	26232 Rs./ Month	Monthly Cost Consumption	17544 Rs./ Month
Yearly Cost Consumption	314784 Rs./ Year	Yearly Cost Consumption	210528 Rs./ Year



Validation of Permanent Corrective Actions

Tangible Benefits

- Yearly Power Saving of 13,032 Units.
- Cost saving of 1,04,256 Rs/ Year.

In-tangible Benefits

- 40 HP motor used as Spare motor for Waste pulper stirrer motor.
- Morale increased due to reduction in power consumption.

Standardization & Horizontal Deployment

- Training on Standard Operating Procedures are given to all the maintenance people for regular inspection & maintenance.


MAY - 2019
PUMP CHECK LIST

Kaizen Sustenance:

WHAT TO DO	Regular Inspection and Maintenance.
HOW TO DO	Regular Training and Standardized Check sheet.
FREQUENCY	Weekly.

Horizontal Deployment:

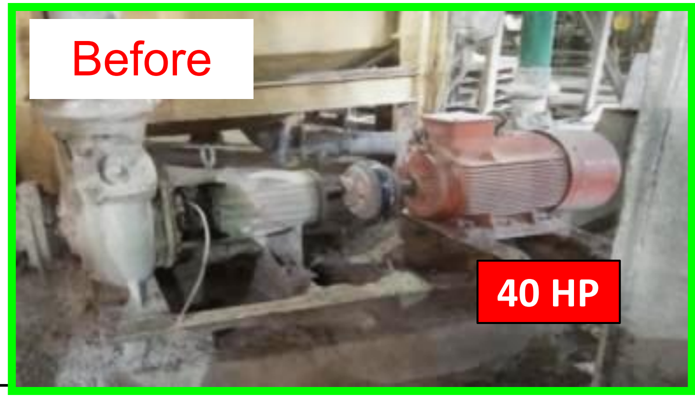
BW	CW	KW	LW	PW1	PW2	SW

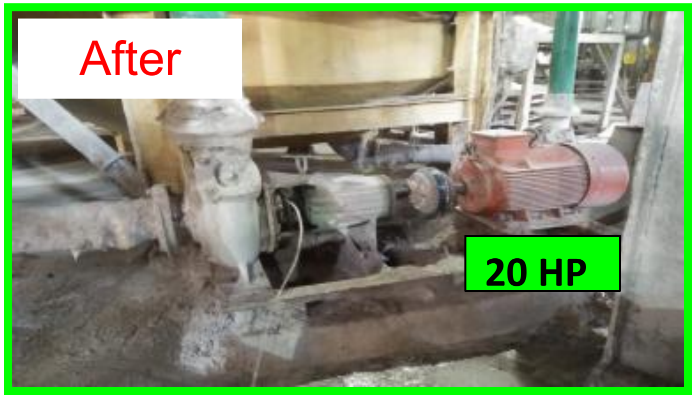
Kaizen Sheet		Category : 02	Company	MM/YY	Kaizen S No
Reactive	Proactive	Innovative √	EVEREST	12/18	2

Kaizen Title: To reduce power consumption of Main Mixer pump motor.

Implemented Area: Shop Floor

Problem/Present Status:
High Power consumption of Main Mixer Pump motor by using pump Motor capacity of 40HP.





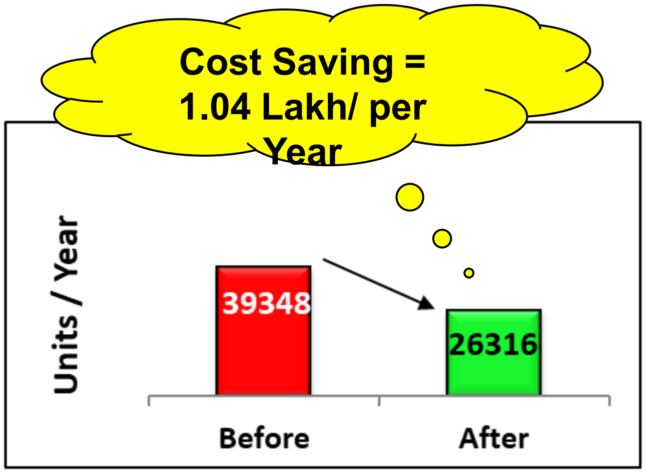
Implemented by:
K. Karthikeyan, M. Vijay Kumar, K. Senthil Kumar, Sasidharan

Root Cause Identification:

- Problem: •More power consumption by Main Mixer Pump motor.
- Why 1: •We are using High capacity Pump Motor of 40Hp.
- Why 2: • Discharge pipe line routing has pipe has 4 Bends in b/w & length of the pipe line is more.
- Why 3: • Design Constraint of Delivery pipe line system.

Result/Benefit:
Qualitative
Proper usage of resources.

Quantitative



Root cause- Design constraint of the Delivery pipe line.

Standardization:
Preventive maintenance check sheet modified.

Idea to eliminate Root Cause To modify pipeline routing arrangement for easy flow of material in pipeline.

Horizontal Deployment

BW	CW	KW	LW	PW	SW

Counter-measure-: Modified pipeline routing for easy flow of material in pipe line. So that, load on the motor is reduced.

Thank You